Introduction

Retrieving reliable evidence from long documents is an important challenge in document question answering. Reports, academic papers, financial statements, and user manuals can contain information spread across many pages. A system must identify relevant evidence before it can produce a dependable answer.

For this project, I used the CRISP-DM methodology to investigate document retrieval in the context of RAG-Anything, a framework designed for multimodal retrieval-augmented generation. The evaluation used questions and evidence annotations from the MMLongBench-Doc dataset. An important distinction is that this project evaluated a **local TF-IDF retrieval baseline**, not the complete RAG-Anything pipeline. The baseline provides a starting point for examining whether relevant document pages can be retrieved for a question.

Dataset and Research Question

Research Question: Can a simple TF-IDF method find the right pages in long documents to help answer questions?

MMLongBench-Doc was the evaluation dataset. It had 1,091 questions across seven fields, plus 135 PDF documents.

The documents included academic papers, financial reports, tutorials, and manuals. Because answers can appear in text, tables, charts, or figures, retrieving the right evidence is challenging.

- Dataset Source: MMLongBench-Doc on Hugging Face
- Research Paper: MMLongBench-Doc
- Framework: RAG-Anything on GitHub

Following the CRISP-DM Process

1. **Business Understanding**
   The practical goal was to examine whether a retrieval system could locate useful evidence in long documents. A retrieval baseline is relevant because a question-answering system cannot reliably ground its answers if it retrieves unrelated pages.

2. **Data Understanding**
   I checked the dataset for structure, missing values, answer types, document categories, and duplicates. It had 1,091 records with no missing fields. One duplicated question had conflicting answers—one gave a number with evidence, the other said “not answerable.” I kept both and noted the issue instead of changing the data.

   I then created a 30-question sample using a fixed random seed and chose five questions for manual testing. These covered a logo color, a financial ratio, a figure-based research question, a political survey, and a monitor safety instruction.

3. **Data Preparation**
   The five selected PDF files were downloaded and their text extracted page by page. I used PyMuPDF for extraction and scikit-learn's TF-IDF vectorizer to represent document pages and questions numerically.

   The TF-IDF representation used lowercase text, English stop-word removal, unigrams and bigrams, and a maximum vocabulary size of 50,000 features. Cosine similarity was used to rank the most relevant pages for each question.

4. **Modeling**
   For each question, the system calculated similarity scores against the indexed PDF pages and returned the three highest-ranked results. This approach is lightweight and reproducible, but it primarily relies on lexical overlap. It does not perform multimodal reasoning, and it may struggle with information contained in images, charts, or complex tables.

5. **Evaluation**
   The five pilot questions were checked by looking at the pages the system retrieved. Four questions had evidence pages, but the line-plot question didn't, so it wasn't included in the hit-rate results.

   Out of those four questions, two had an exact page match **(50%)**, and three had a page that was next to the correct one **(75%)**. These numbers are only early estimates because the page numbering in the dataset and the PDFs still needs to be cleaned up.

   The monitor-manual question was the clearest success because the retrieved page showed the correct safety instruction. The other questions were less clear. Even when the system found the right PDF, the part didn't always show enough evidence to answer the question.

   Since only five questions were tested, this should be seen as a small pilot, not a reliable measure of overall performance.

6. **Deployment & Delivery**
   The project includes the Colab notebook, pilot sample, TF-IDF results, manual-review notes, and the written report. These files make the experiment easy to reproduce.

   The full RAG-Anything system wasn't run or deployed, so this work should be seen as a CRISP-DM study of a simple document-retrieval baseline—not a test of the full multimodal framework.

Limitations

1. Only five questions were checked
2. TF-IDF can miss answers when wording changes
3. Extracted text loses charts and tables
4. Some annotations were unclear, page numbering needs to be standardized
5. The full RAG-Anything system wasn't tested

The results show why you need both numbers and manual review. Finding the right PDF isn't enough—the retrieved page has to actually contain evidence for the answer.

Conclusion

CRISP-DM helped keep the project organized. The TF-IDF baseline worked clearly for one question, but the other questions showed problems with page matching, missing visuals, and unclear annotations.

Next steps are to fix page numbering, test more questions, make a clearer scoring guide, and compare this baseline with the full RAG-Anything system. This will show whether multimodal retrieval actually improves question answering.

Project Artifacts

The project notebook, reports, pilot sample, retrieval results, and manual-review records are available in the CRISP-DM subdirectory of my repository: https://github.com/Olivia883/Cmpe255-data-mining-process/tree/main.